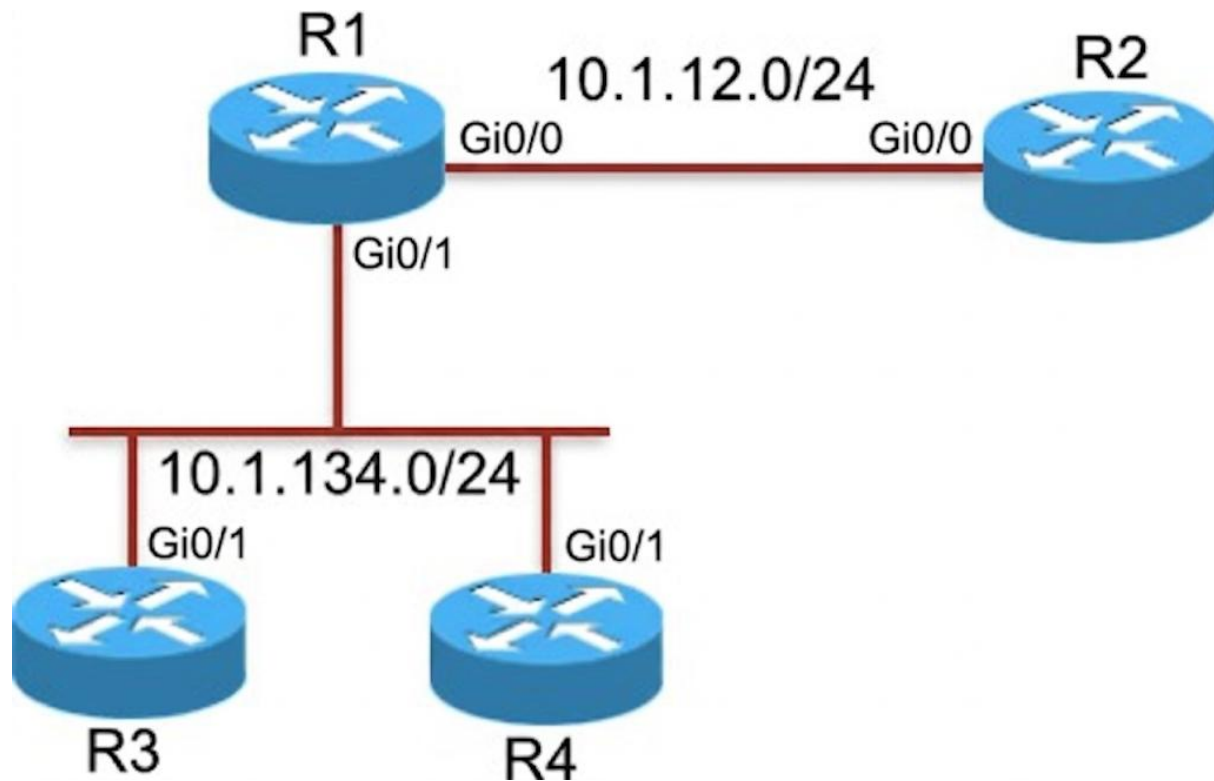


Standard ACL

Lab Overview & Objectives



Tasks

- Configure hostname and IP addresses on all routers using GigabitEthernet interfaces.
- Configure OSPF (Process ID 1, Area 0) on all routers to establish routing.
- Configure standard ACL on R1 to deny R3 from reaching R2.
- Remaining traffic should not be denied due to this ACL.

Key Concepts

Access Control List (ACL) is a packet filtering technology that allows us to filter IP traffic based on Layer 3 and Layer 4 information. It can inspect source/destination port numbers and IP addresses. There are two primary types of ACLs: Standard ACLs and Extended ACLs.

Standard ACL is used to filter packets based solely on the source IP information. It doesn't provide flexibility to define complex access list rules, as it doesn't support source/destination port information or destination IP addresses in the Access List Entry (ACE). On Cisco IOS, standard ACLs use numbers between **1 and 99** (and extended range 1300–1999).

In this lab, it is necessary to enable routing before configuring the access-list on R1. Once end-to-end reachability is verified, the ACL is configured as required. When applying an access-list, options exist for inbound and outbound filtering. In this scenario, it is applied **outbound**, adhering to the best practice that standard ACLs should generally be applied as close to the destination as possible.

Router Configurations

Router R2

```
enable
```

```
configure terminal
```

```
hostname R2
```

```
!
```

```
interface GigabitEthernet0/0
```

```
ip address 10.1.12.2 255.255.255.0
```

```
no shutdown
```

```
!
```

```
router ospf 1
```

```
network 10.0.0.0 0.255.255.255 area 0
```

```
!
```

```
end
```

Router R1 (ACL Location)

```
enable
```

```
configure terminal
```

```
hostname R1
```

```
!
```

```
interface GigabitEthernet0/0
```

```
ip address 10.1.12.1 255.255.255.0

no shutdown

!

interface GigabitEthernet0/1

ip address 10.1.134.1 255.255.255.0

no shutdown

!

router ospf 1

network 10.0.0.0 0.255.255.255 area 0

!

access-list 1 deny 10.1.134.3

access-list 1 permit any

!

interface GigabitEthernet0/0

ip access-group 1 out

!

end
```

Router R3

```
enable

configure terminal

hostname R3

!

interface GigabitEthernet0/1

ip address 10.1.134.3 255.255.255.0
```

```
no shutdown
```

```
!
```

```
router ospf 1
```

```
network 10.0.0.0 0.255.255.255 area 0
```

```
!
```

```
end
```

Router R4

```
enable
```

```
configure terminal
```

```
hostname R4
```

```
!
```

```
interface GigabitEthernet0/1
```

```
ip address 10.1.134.4 255.255.255.0
```

```
no shutdown
```

```
!
```

```
router ospf 1
```

```
network 10.0.0.0 0.255.255.255 area 0
```

```
!
```

```
end
```

Verification & Testing

Once OSPF and the ACL are configured, verify the routing information before testing end-to-end reachability.

1. Check OSPF Routing Table on R3

t

R3#show ip route ospf

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP

D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area

...

Gateway of last resort is not set 10.0.0.0/8 is variably subnetted, 3 subnets, 2 masks

O 10.1.12.0/24 [110/2] via 10.1.134.1, 00:05:12, GigabitEthernet0/1

2. Connectivity Tests

Test ICMP reachability from both ****R3**** and ****R4**** to ****R2 (10.1.12.2)****. The ping packet from ****R3**** should fail (denied by ACL), while ****R4**** should successfully reach the destination.

Ping from R3 (Expected Failure)

``text

R3#ping 10.1.12.2

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 10.1.12.2, timeout is 2 seconds:

U.U.U

Ping from R4 (Expected Success)

R4#ping 10.1.12.2

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 10.1.12.2, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 1/2/4 ms

3. Verify ACL Matches on R1

R1#show access-lists

Standard IP access list 1

10 deny 10.1.134.3 (5 matches)

20 permit any (5 matches)

Summary Analysis

In the verification outputs above:

- R3 is unable to ping 10.1.12.2 because the standard ACL on R1 denies traffic originating from 10.1.134.3 before it exits GigabitEthernet0/0.
- The show access-lists output confirms 5 packet matches on the deny entry corresponding to R3's ping attempts.
- R4 (10.1.134.4) successfully pings the destination, matching the permit any rule, which is also confirmed by 5 packet matches on statement 20.